

- ① Perform the following operations in LIFO data structures having an array size of 5. Push(45), Push(33), Push(22), Push(11), Push(77), Pop(), Push(44), Pop(), Pop(), Pop(). Print the peak element of data structure and its index position. with a neat sketch explained in detail.

Size = 5

top = -1

Push (45)

4	
3	
2	
1	
0	45

top = top + 1

= -1 + 1

top = 0

← top = 0
peak element = 45.

Push (22)

4	
3	
2	
1	22
0	45

top = top + 1

= 0 + 1

top = 1

← top = 1
peak element = 22.

Push (11).

0	1	2	3	4
45	22	11		

↑

top = 2

Peak element = 11

top = 1 + 1

top = 2

Push (77)

0	1	2	3	4
45	22	11	77	

↑

top = 3

Peak element = 77

top = 2 + 1

top = 3.

Pop()

0	1	2	3	4
45	22	11		

↑

top = 2

Peak = 11

top = top - 1

= 3 - 1

top = 2.

Pop()

0	1	2	3	4
45	22	11		

↑

top = 2

Peak = 11

Push (44)

0	1	2	3	4
45	22	11	44	

↑

top = 3

Peak = 44

top = 2 + 1

top = 3.

Pop()

0	1	2	3	4
45	22			

↑

top = 1

Peak = 22.

Push (33)

0	1	2	3	4
45	22	11	44	33

↑

top = 4

Peak = 33.

Pop()

0	1	2	3	4
45	22	11	44	

↑

top = 3

Peak = 44.

∴ The peak element is 22 and its index position is 1.

2. Perform the following operations in queue using a size of 5.
 enqueue(20), enqueue(10), enqueue(30), dequeue(),
 enqueue(80), dequeue(), enqueue(90), enqueue(100),
 dequeue(), dequeue(), enqueue(60), enqueue(90)
 finally display the front and rear elements with its positions.

Enqueue (20)

front = rear = 0.

0	1	2	3	4
20				

Enqueue (10)

front = 0 and rear = 1

0	1	2	3	4
20	10			

Enqueue (30)

front = 0, rear = 2

0	1	2	3	4
20	10	30		

Dequeue ()

front = 1, rear = 2.

0	1	2	3	4
	10	30		

Enqueue (80)

front = 1, rear = 3.

0	1	2	3	4
	10	30	80	

dequeue ()

front = 2, rear = 3

		30	80	
--	--	----	----	--

Enqueue (90)

front = 2, rear = 4

		30	80	90
--	--	----	----	----

Enqueue (100)

front = 2, rear = 0

100		30	80	90
-----	--	----	----	----

Dequeue ()

front = 3 and rear = 0

100			80	90
-----	--	--	----	----

Dequeue ()

front = 4 and rear = 0

100				90
-----	--	--	--	----

Enqueue (60)

front = 4 rear = 1

100	60			90
-----	----	--	--	----

∴ The front = 4

The rear = 2.

3. Convert the following infix expression into postfix.

$$B - (C/D + (E * F * G) / H)^* J$$

Scanning

Stack

Post fix
Expression.

B

[]

B

-

[-]

B

C

[- | C |]

B

C

[- | C | C |]

BC

/

[- | C | / |]

BC/

D

[- | C | / | D |]

BCD

+

[- | C | + |]

BCD/

C

[- | C | + | C |]

BCD/.

E

[- | C | + | C | E |]

BCD/E

.

[- | C | + | C | . |]

BCD/E

F

[- | C | + | C | . | F |]

BCD/E

*

[- | C | + | C | . | * |]

BCD/E F .

G

[- | C | + | C | . | * | G |]

BCD/E F . G^{*}

)

[- | C | + |]

BCD/E F . G^{*}

*

[- | * |]

BCD/E F . G^{*} H | + J

J

[- | * | J |]

BCD/E F . G^{*} H | + J^{*}

End.

Evaluate Postfix expression.

a) $9384 / * +$

Scanning

stack

operation.

9

[9]

3

[9 | 3]

8

[9 | 3 | 8]

4

[9 | 3 | 8 | 4]

/

[9 | 3 | 2]

$8 / 4$

$3 * 2$

*

[9 | 6]

$9 + 6$

b.

$62 + 63 / *$

Scanning

stack

operation.

6

[6]

2

[6 | 2]

+

[8]

6

[8 | 6]

3

[8 | 6 | 3]

/

[8 | 2]

*

[16]

$\therefore$ The answer is 16.